Project 96: Multi-Criteria Library Resource Distribution

Python Mini Project - College Assignment

1. Aim

To distribute available library books among branches according to borrowing demand and assigned priority, and calculate unmet demand.

2. Algorithm

• Start the program.

• Store book names, stock, branch names, demand, and priority.

• Process branches in ascending priority number (1 is highest).

• For each book, allocate the smaller of available stock and demand.

• Subtract allocated copies from stock and calculate unmet demand.

• Display allocations, remaining stock, and total unmet demand.

• Stop.

3. Python Program

```
# Project 96: Multi-Criteria Library Resource Distribution
print("=====================================")
print("     LIBRARY RESOURCE DISTRIBUTION")
print("=====================================")

books = ["Python", "Mathematics", "Physics"]
stock = [15, 10, 8]
branches = ["Central", "Science", "Engineering"]
priority = (1, 2, 3)  # 1 = highest priority

demand = [
    [8, 5, 4],
    [6, 4, 3],
    [5, 6, 4]
]
allocated = [[0, 0, 0], [0, 0, 0], [0, 0, 0]]
unmet = [[0, 0, 0], [0, 0, 0], [0, 0, 0]]

print("\nAVAILABLE BOOK STOCK")
for i in range(len(books)):
    print(books[i], ":", stock[i])

print("\n=====================================")
print("BOOK ALLOCATION REPORT")
print("=====================================")

for p in range(1, 4):
    for i in range(len(branches)):
        if priority[i] == p:
            print("\nBranch:", branches[i])
            print("Priority:", priority[i])
            for j in range(len(books)):
                required = demand[i][j]
                if stock[j] >= required:
                    allocated[i][j] = required
                else:
```

```
                    allocated[i][j] = stock[j]
                stock[j] = stock[j] - allocated[i][j]
                unmet[i][j] = required - allocated[i][j]
                print("\nBook:", books[j])
                print("Demand:", required)
                print("Allocated:", allocated[i][j])
                print("Unmet Demand:", unmet[i][j])
print("\n=====================================")
print("REMAINING BOOK STOCK")
print("=====================================")
for i in range(len(books)):
    print(books[i], ":", stock[i])

print("\n=====================================")
print("TOTAL UNMET DEMAND")
print("=====================================")
total_unmet = 0
for i in range(len(branches)):
    branch_unmet = 0
    for j in range(len(books)):
        branch_unmet += unmet[i][j]
    total_unmet += branch_unmet
    print(branches[i], ":", branch_unmet)
print("\nTotal Unmet Demand:", total_unmet)
print("\n=====================================")
print("       DISTRIBUTION COMPLETED")
print("=====================================")
```

4. Sample Output (from running the program)

```
=====================================
     LIBRARY RESOURCE DISTRIBUTION
=====================================

AVAILABLE BOOK STOCK
Python : 15
Mathematics : 10
Physics : 8

=====================================
BOOK ALLOCATION REPORT
=====================================

Branch: Central
Priority: 1

Book: Python
Demand: 8
Allocated: 8
Unmet Demand: 0

Book: Mathematics
Demand: 5
Allocated: 5
Unmet Demand: 0

Book: Physics
Demand: 4
Allocated: 4
Unmet Demand: 0

Branch: Science
Priority: 2

Book: Python
Demand: 6
```

```
Allocated: 6
Unmet Demand: 0

Book: Mathematics
Demand: 4
Allocated: 4
Unmet Demand: 0

Book: Physics
Demand: 3
Allocated: 3
Unmet Demand: 0

Branch: Engineering
Priority: 3
Book: Python
Demand: 5
Allocated: 1
Unmet Demand: 4

Book: Mathematics
Demand: 6
Allocated: 1
Unmet Demand: 5

Book: Physics
Demand: 4
Allocated: 1
Unmet Demand: 3

====================================
REMAINING BOOK STOCK
====================================
Python : 0
Mathematics : 0
Physics : 0

====================================
TOTAL UNMET DEMAND
====================================
Central : 0
Science : 0
Engineering : 12

Total Unmet Demand: 12

====================================
       DISTRIBUTION COMPLETED
====================================
```

5. Python Concepts Used

Concept	Application
Variables and data types	Store stock, demand, and shortages
Operators and conditions	Compare demand with stock
Loops	Process branches and books
Lists and nested lists	Store demand and allocations
Tuples	Store priority values

Concept	Application
Strings	Book and branch names

6. Result

The program allocates all 33 available copies. Central and Science branches receive their full requests, while Engineering has 12 copies of unmet demand. Remaining stock is zero for all three book titles.